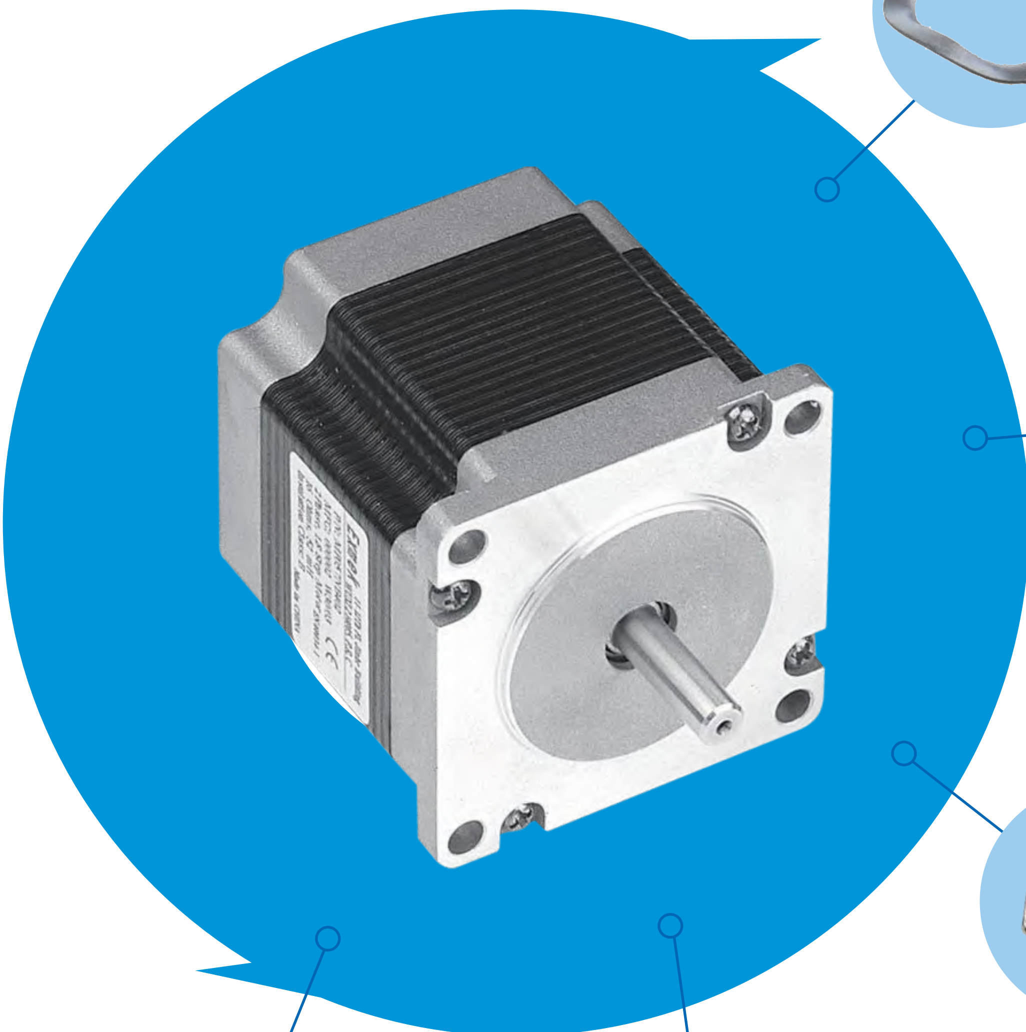
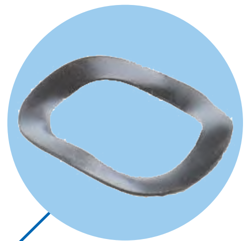
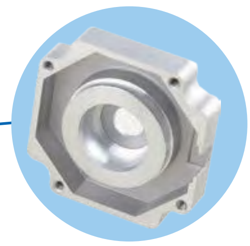
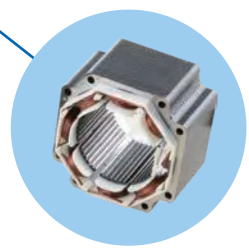
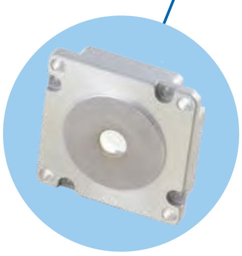
STEPPER MOTOR











MB057NB

We can customize it according to customer's requirements

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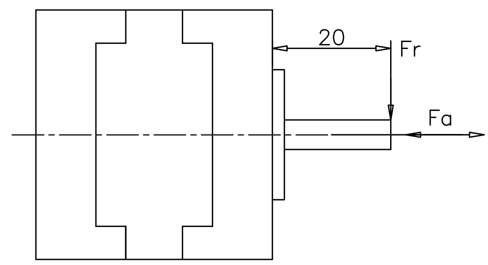
STEPPER MOTOR
General Information

Basic Parameters

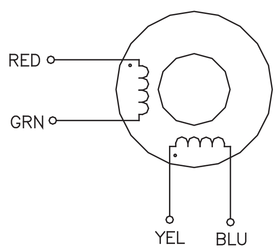
- Step Angle Accuracy: ±5%
- Ambient Temperature: -20°C~+50°C
- Insulation Resistance: 100M Ω Min.
- Dielectric Strength: 500VAC ,1S,2mA
- Insulation Class: B
- Frequancy-Torque cure on request

Max. Axial Force And Radial Force

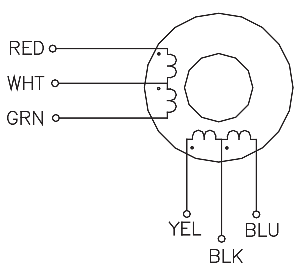
Item	Fr(N)	Fa(N)
Distance from the flange (mm)	20	
MP020 Shaft φ5mm	8	4
MP039;MP042 Shaft φ5mm	20	7
MP057 Shaft φ6.35mm	52	10
MP057 Shaft φ8mm	63	14
MP086 Shaft φ9.5mm	100	25
NP086 Shaft φ14mm	200	25
MP110 Shaft φ19.05mm	240	80



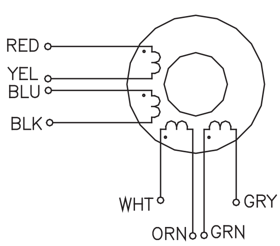
Mode Of Connection



4 LEAD WIRING DIAGRAM				
LEAD WIRE COLOR	RED	GRN	YELLOW	BLUE
BIPOLAR DRIVE	A	A/	B	B/



6 LEAD WIRING DIAGRAM							
LEAD WIRE COLOR		RED	WHITE	GREEN	YELLOW	BLACK	BLUE
BIPOLAR DRIVE	PARRALLEL CONNECTION	A	A/	N/C	B	B/	N/C
	SERIES CONNECTION	A	N/C	A/	B	N/C	B/
UNIPOLAR DRIVE		A	COM	B	C	COM	D



8 LEAD WIRING DIAGRAM									
LEAD WIRE COLOR		RED	BLUE	YELLOW	BLACK	WHITE	GREEN	ORANGE	GRAY
BIPOLAR DRIVE	PARRALLEL CONNECTION	A		A/		B		B/	
	SERIES CONNECTION	A			A/	B			B/
UNIPOLAR DRIVE		A	COM		B	C	COM		D

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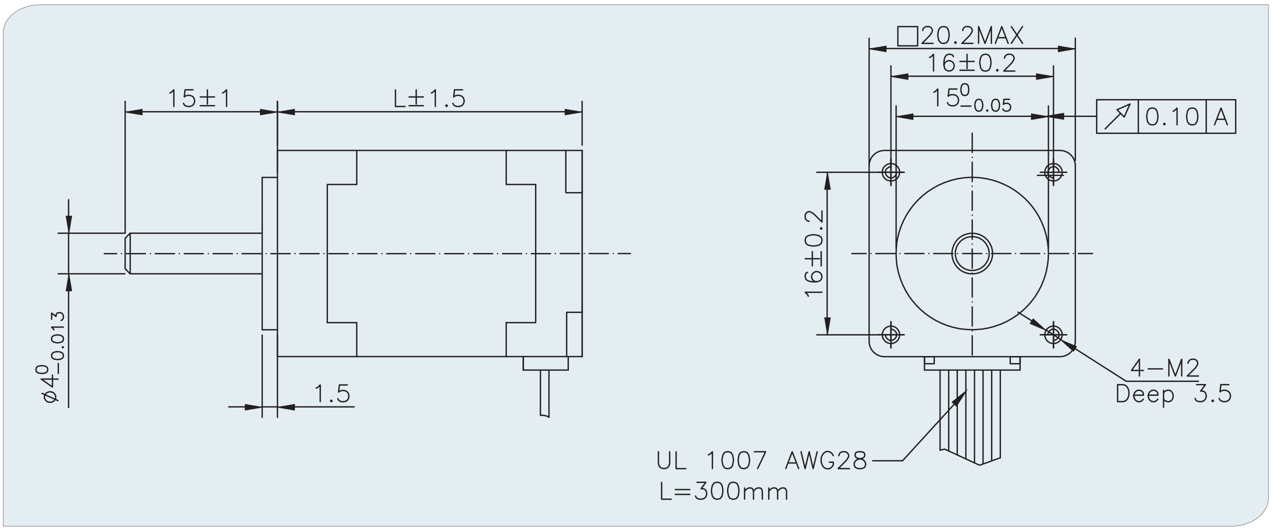
MP020NA
Stepper Motor

- 20mm, 1.8Degree Hybrid Stepper Motor

Mechanical



Specifications

Part No.		MP020NA101	MP020NA300
Phase current	(A)	0.6	0.6
Phase Resistance	(Ohms)	6.5	6.5
Phase Inductance	(mH)	1.7	0.8
Holding Torque	(Ncm)	1.8	2
Detent Torque	(Ncm)	0.2	0.2
No.of Wires		4	6
Rotor Inertia	($g.cm^2$)	2.5	2.9
Length	(mm)	30	33
Weight	(g)	60	60

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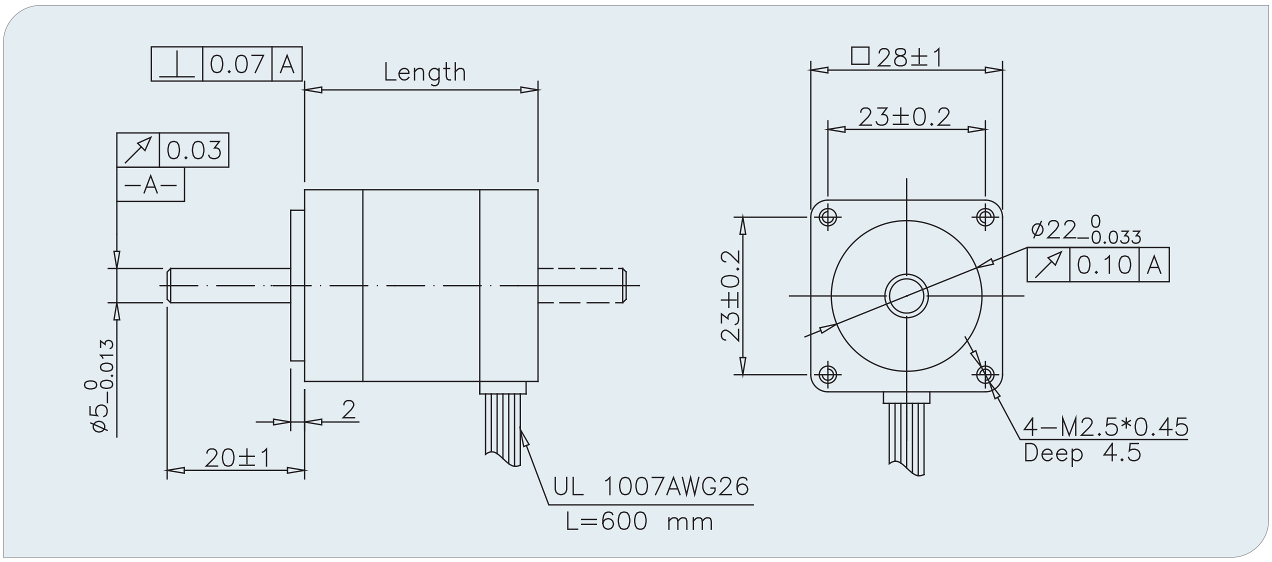
MP028NB
Stepper Motor

- 28mm, 1.8Degree Hybrid Stepper Motor

Mechanical



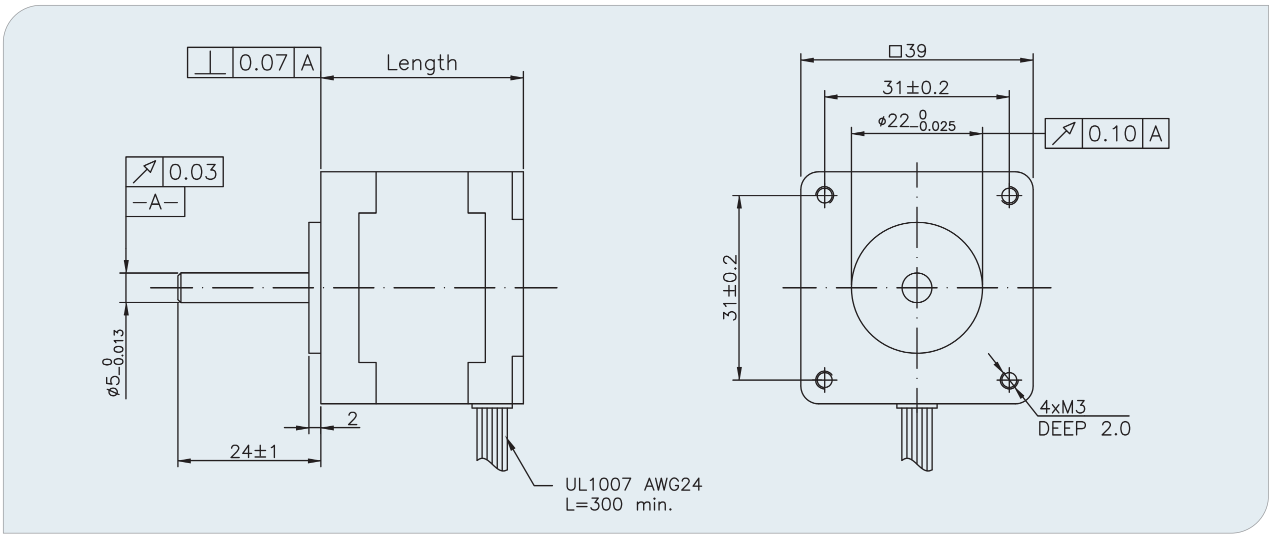
Specifications

Part No.	Phase Current	Phase Resistance	Phase Inductance	Holding Torque	Detent Terque	No. of Wires	Rotor Inertia	Length	Weight
	(A)	(OHMS)	(mH)	(Ncm)	(Ncm)		(g.cm^2)	(mm)	(g)
MP028NB101	0.67	5.6	4.2	6	0.8	4	9.5	31.5	100
MP028NB102	0.47	2.8	1	4	0.8	6	9.5	31.5	100
MP028NB103	1.3	1.4	1.1	6.5	0.8	4	9.5	31.5	100
MP028NB201	0.67	6.8	4.9	9.5	1.2	4	12	44.5	180
MP028NB202	1	3.4	1.2	7.5	1.2	6	12	44.5	180
MP028NB301	0.67	9.2	5.7	12	1.5	4	18	50.5	210
MP028NB302	0.96	4.6	1.8	9	1.5	6	18	50.5	210
MP028NB303	1	4.3	2	9.5	1.5	6	18	50.5	210

MP039NA
Stepper Motor

- 39mm, 1.8Degree Hybrid Stepper Motor

Mechanical



Specifications

Part No.	Phase Current (A)	Phase Resistance (OHMS)	Phase Inductance (mH)	Holding Torque (Ncm)	Detent Torque (Ncm)	No.of Wires	Rotor Inertia ($g.cm^2$)	Length (mm)	Weight (g)
MP039NA101	0.4	6.6	7.5	6.5	1.2	4	11	20	120
MP039NA103	0.28	13	7.5	4.6	0.8	6	11	20	120
MP039NA106	0.6	7	6.4	8.3	1.2	4	11	20	120
MP039NA201	0.4	30	32	20.6	1.8	4	20	34	180
MP039NA203	0.3	40	20	12.7	1.8	6	20	34	180
MP039NA204	0.6	15	16	21.6	1.8	6	20	34	180
MP039NA302	0.5	24	45	28.4	2.16	4	28	38	200
MP039NA303	0.8	7.5	6	19.6	2.16	6	28	38	200
MP039NA401	0.8	40	100	27.4	2.5	4	40	44	250
MP039NA501	0.8	20	35	30	2.5	4	60	53	320

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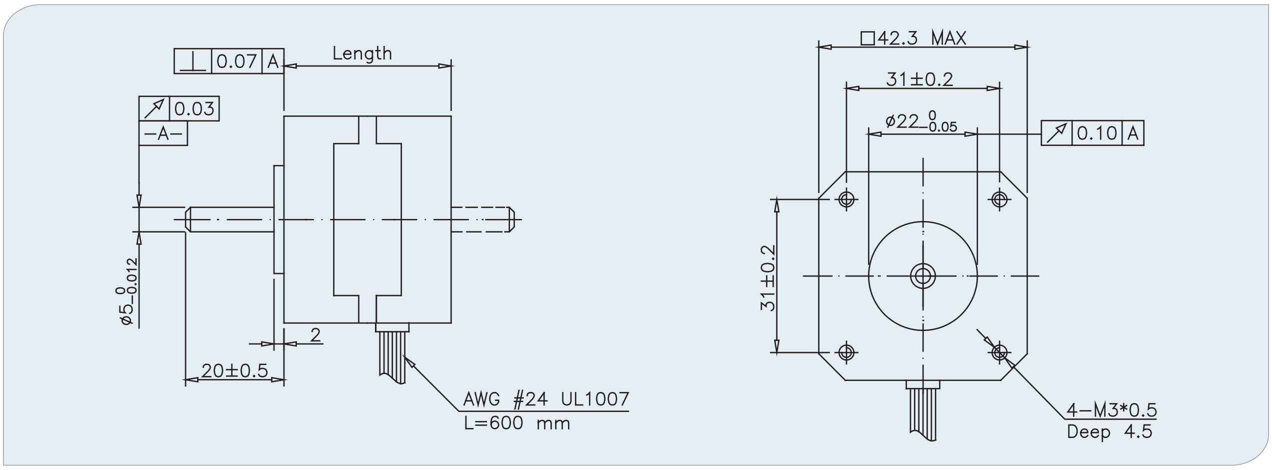
MP042NB

Stepper Motor

- 42mm, 1.8Degree Enhanced Hybrid Stepper Motor

Mechanical



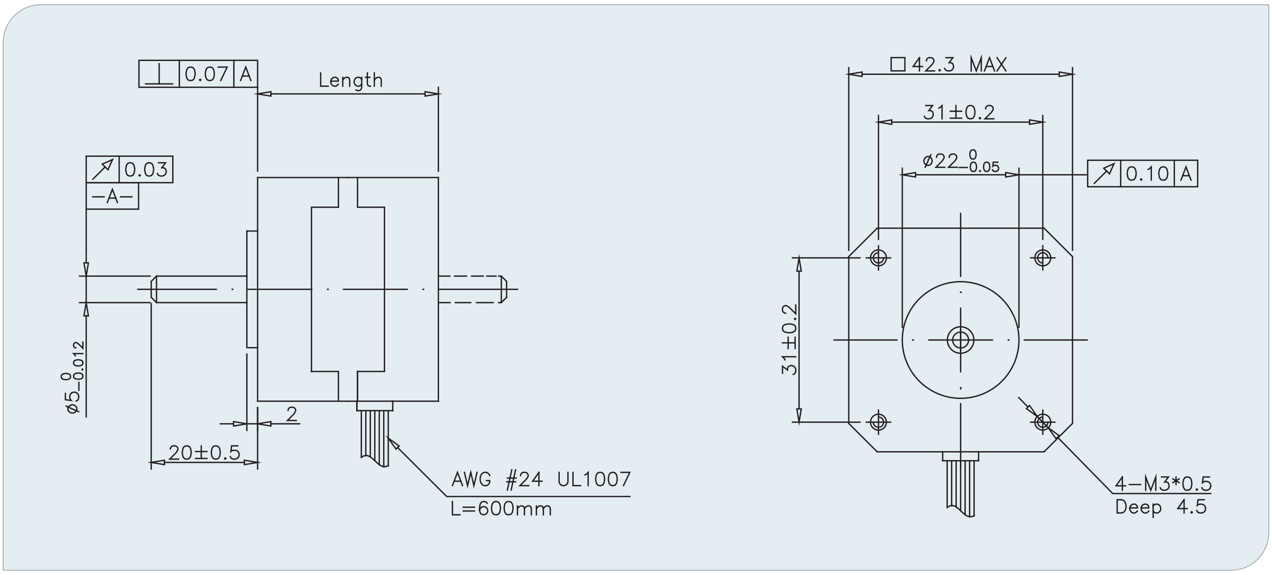
Specifications

Part No.	Phase Current (A)	Phase Resistance (OHMS)	Phase Inductance (mH)	Holding Torque (Ncm)	Detent Torque (Ncm)	No.of Wires	Rotor Inertia (g.cm²)	Length (mm)	Weight (g)
MP042NB001	0.4	24	3.6	17	2	4	20	24	150
MP042NB002	0.6	6.6	8.5	12	1.42	4	20	24	150
MP042NB111	0.6	12	15	25.5	1.96	4	34	34	200
MP042NB123	0.4	30	40	24.5	1.96	4	34	34	200
MP042NB136	0.28	44	38	16.7	1.96	6	34	34	200
MP042NB140	1	3.6	3.0	19.6	1.96	6	34	34	200
MP042NB302	1.2	3	3	26.5	2.16	6	54	40	240
MP042NB316	3.5	0.65	1.2	50	2.74	4	54	40	240
MP042NB335	1.7	1.5	3.2	44	2.2	4	54	40	240
MP042NB355	0.9	5	11.5	39.3	2.94	4	54	40	350
MP042NB519	1	4.6	4	33.3	2.74	6	68	48	340
MP042NB522	0.5	22	33	50	2.74	4	68	48	340
MP042NB523	1.5	3.15	6.15	49	2.74	4	68	48	340
MP042NB530	3	0.63	1.03	50	2.2	4	68	48	340
MP042NB531	2.5	1.25	1.8	48	2.8	4	68	48	340
MP042NB603	0.86	4.94	11.45	70	3.5	4	82	60	480
MP042NB606	1.2	12	35	75	6	4	82	60	480
MP042NB607	1.2	6	15.6	80	3.5	4	82	60	480

MP042SB
Stepper Motor

- 42mm, 0.9Degree Enhanced Hybrid Stepper Motor

Mechanical



Specifications

Part No.	Phase Current	Phase Resistance	Phase Inductance	Holding Torque	Detent Torque	No.of Wires	Rotor Inertia	Length	Weight
	(A)	(OHMS)	(mH)	(Ncm)	(Ncm)		(g.cm^2)	(mm)	(g)
MP042SB001	1	8.7	16.5	14.5	1.5	4	20	22	150
MP042SB004	1	9.2	21	16.5	1.5	4	20	22	150
MP042SB104	1.2	2.1	4.9	20.6	1.96	4	34	34	220
MP042SB105	0.5	20	35	18.6	2.16	4	34	34	220
MP042SB110	0.67	8.4	16	21.2	1.5	6	34	34	220
MP042SB301	1.2	3.0	4.0	29.4	2.45	6	54	40	280
MP042SB302	0.8	7.5	8.6	25.5	2.45	6	54	40	280
MP042SB303	0.4	30	36	25.5	2.45	6	54	40	280
MP042SB304	1.68	1.5	4.0	35.3	2.45	4	54	40	280
MP042SB501	1.2	3.3	4.5	31.4	2.74	6	68	48	350
MP042SB502	0.8	7.5	6.3	31.4	2.74	6	68	48	350
MP042SB503	0.4	30	31	31.4	2.74	6	68	48	350
MP042SB504	1.68	1.54	3.5	41.2	2.74	4	68	48	350
MP042SB509	1	5	8	49	2.74	4	68	48	350

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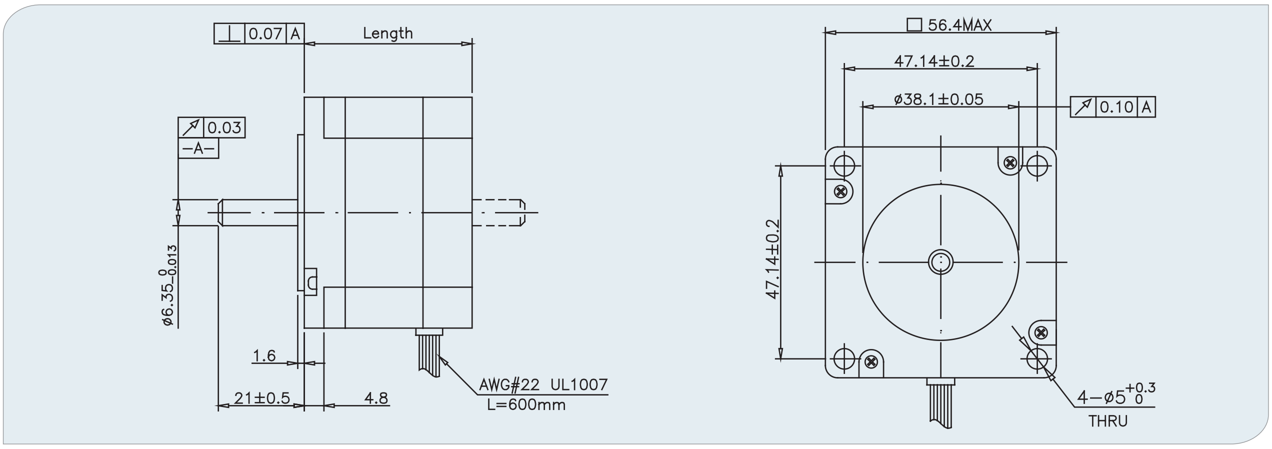
MP057NB

Stepper Motor

- 57mm, 1.8Degree Enhanced Hybrid Stepper Motor

Mechanical



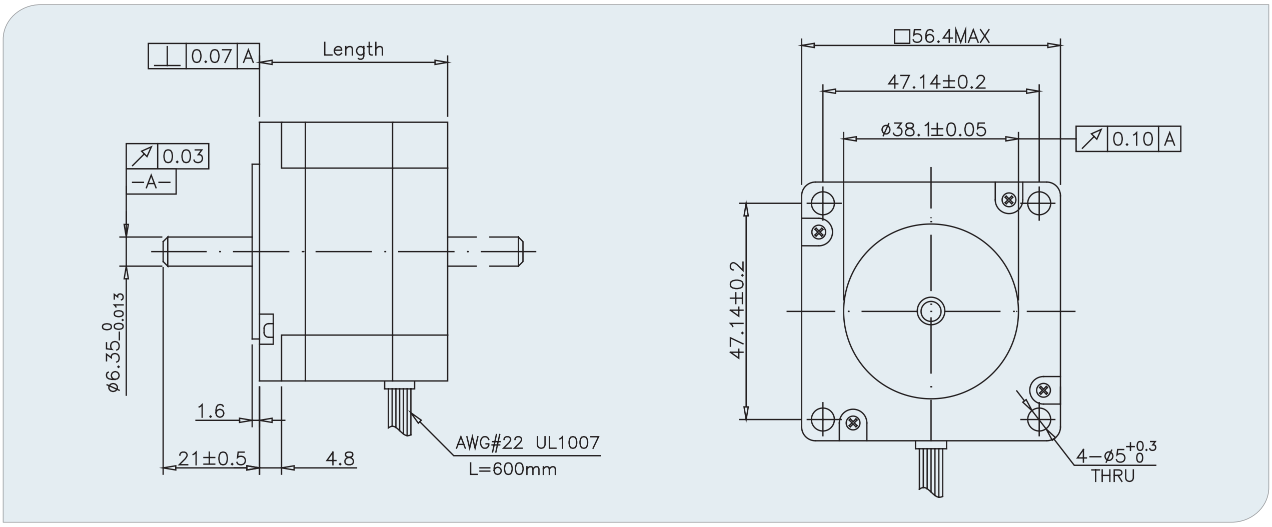
Specifications

Part No.	Phase Current	Phase Resistance	Phase Inductance	Holding Torque	Detent Torque	No.of Wires	Rotor Inertia	Length	Weight
	(A)	(OHMS)	(mH)	(Ncm)	(Ncm)		(g.cm^2)	(mm)	(g)
MP057NB010	1.3	3	5	41	2.6	4	135	41	420
MP057NB020	1.3	2.7	5.6	50	4	4	135	41	420
MP057NB024	2	1.2	3.2	55	4	4	135	41	420
MP057NB201	1	6.6	9	70.6	5.88	6	275	50	550
MP057NB204	1	5.5	16.5	83.3	5.88	4	275	50	550
MP057NB208	2	1.8	5.2	80	4	4	275	50	550
MP057NB222	2.8	0.83	2.2	100	3.6	4	275	50	650
MP057NB303	2	1.8	2.5	88.2	8.33	6	300	56	600
MP057NB316	4.2	0.35	0.66	110	6.3	6	300	56	700
MP057NB331	1.4	3.6	10.8	117	3.92	4	300	56	700
MP057NB417	2.1	2	6.5	161	6.8	4	480	78	1000
MP057NB420	1.4	4.5	14.4	175	6.8	4	480	78	1000
MP057NB423	2.5	1.5	4.4	150	7	4	480	78	1000
MP057NB431	5	0.25	0.95	140	7.85	4	480	78	1100
MP057NB490	4.2	0.5	2.2	170	6.8	4	480	78	1100
MP057NB504	2	3.8	10	248	12	4	800	115	1500
MP057NB505	2.7	2	5.5	220	12	4	800	115	1500
MP057NB513	2	7	32	300	12	4	800	115	1500
MP057NB522	6	0.5	2	280	12	4	800	115	1600

MP057SB
Stepper Motor

- 57mm, 0.9Degree Enhanced Hybrid Stepper Motor

Mechanical



Specifications

Specifications	Phase Current	Phase Resistance	Phase Inductance	Holding Torque	Detent Torque	No.of Wires	Rotor Inertia	Length	Weight
	(A)	(OHMS)	(mH)	(Ncm)	(Ncm)		(g.cm^2)	(mm)	(g)
MP057SB101	1	5.7	7	19.6	3.92	6	135	41	420
MP057SB102	2	1.4	2	19.6	3.92	6	135	41	420
MP057SB103	3	0.73	0.86	24.5	3.92	6	135	41	420
MP057SB104	2.8	0.8	1.9	29.4	3.92	4	135	41	420
MP057SB201	1	7.4	17	78.4	5.8	6	300	56	600
MP057SB202	2	1.8	5.2	88.2	5.8	6	300	56	600
MP057SB203	3	0.75	1.1	88.2	5.8	6	300	56	600
MP057SB204	2.8	0.9	3.3	98	5.8	4	300	56	600
MP057SB206	2.2	1.5	7.6	120	6.3	4	300	56	600
MP057SB301	1	8.6	14	127.4	10	6	480	78	1000
MP057SB302	2	2.3	6.7	137.2	10	6	480	78	1000
MP057SB303	2.8	1.13	6.1	147	10	4	480	78	1000
MP057SB501	2	6	32	250	12	4	800	115	1300

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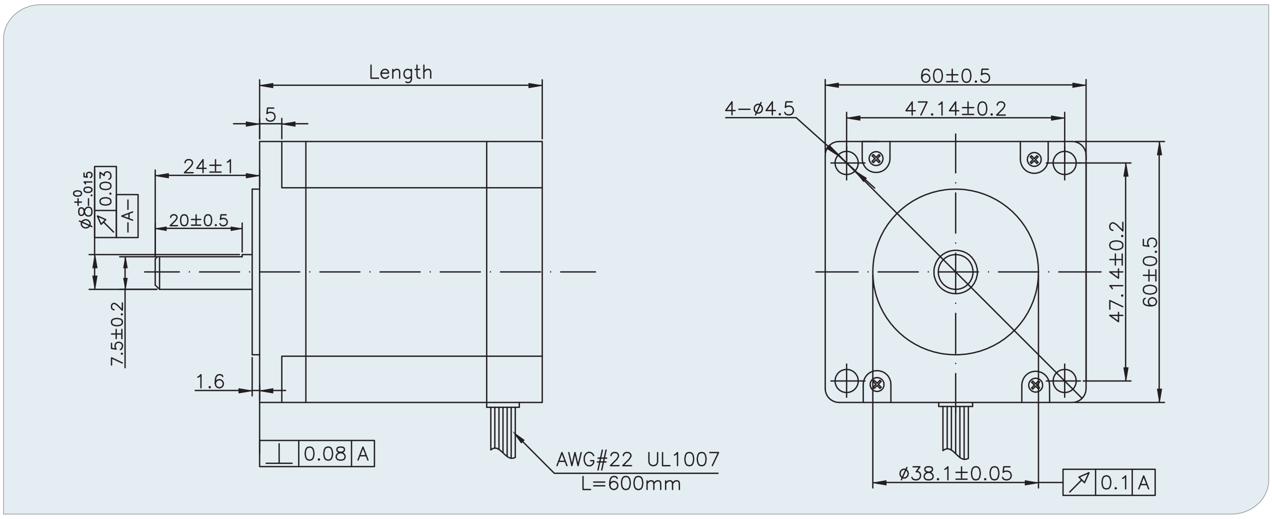
MP060NB

Stepper Motor

- 60mm, 1.8Degree Enhanced Hybrid Stepper Motor

Mechanical



Specifications

Part No.	Connection	Phase Current (A)	Phase Resistance (OHMS)	Phase Inductance (mH)	Holding Torque (Ncm)	Detent Torque (Ncm)	No.of Wires	Rotor Inertia (g.cm^2)	Length (mm)	Weight (Kg)
MP060NB100	Parallel	2.8	0.75	2	110	6	8	275	45	0.6
	Series	1.4	3	8	110	6	8	275	45	0.6
	Unipolar	2	1.5	2	78	6	8	275	45	0.6
MP060NB200	Parallel	2.8	0.9	3.6	165	9	8	300	56	0.77
	Series	1.4	3.6	14.4	165	9	8	300	56	0.77
	Unipolar	2	1.8	3.6	117	9	8	300	56	0.77
MP060NB300	Parallel	2.8	1.2	4.6	210	12	8	570	65	1
	Series	1.4	4.8	18.4	210	12	8	570	65	1
	Unipolar	2	2.4	4.6	150	12	8	570	65	1
MP060NB400	Parallel	2.8	1.2	5.6	270	15	8	579	77	1.2
	Series	1.4	4.8	22.4	270	15	8	579	77	1.2
	Unipolar	2	2.4	5.6	190	15	8	579	77	1.2
MP060NB500	Parallel	2.8	1.5	6.8	310	17	8	840	86	1.4
	Series	1.4	6	27.2	310	17	8	840	86	1.4
	Unipolar	2	3	6.8	220	17	8	840	86	1.4

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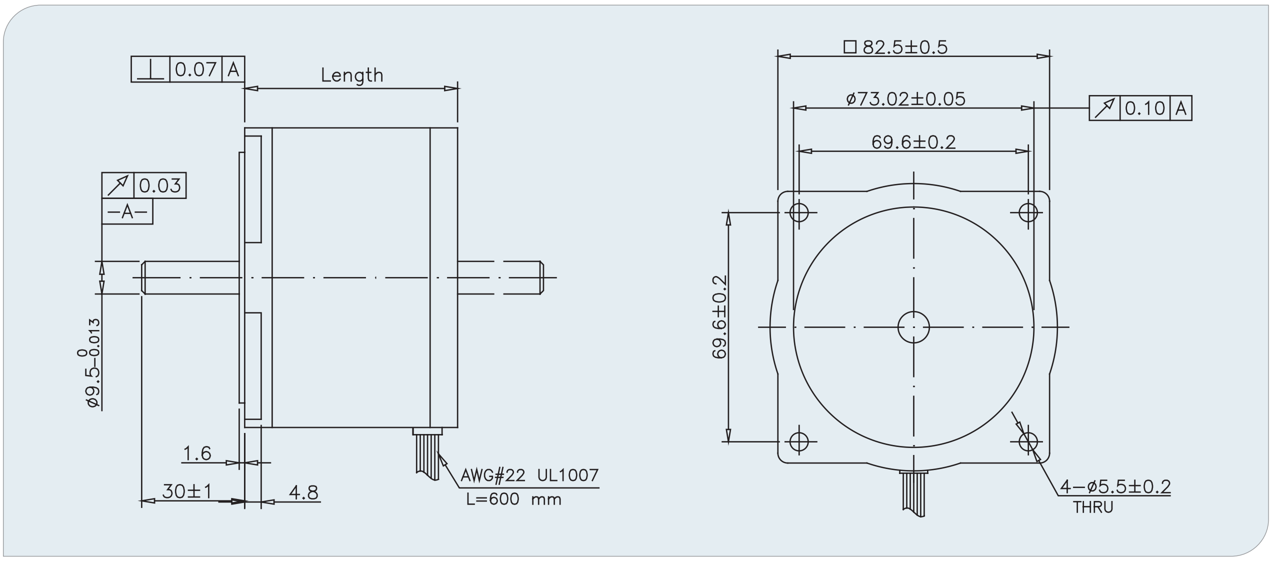
MP086NA

Stepper Motor

- 86mm, 1.8Degree Enhanced Hybrid Stepper Motor

Mechanical



Specifications

Part No.	Phase Current	Phase Resistance	Phase Inductance	Holding Torque	Detent Torque	No.of Wires	Rotor Inertia	Length	Weight
	(A)	(OHMS)	(mH)	(Ncm)	(Ncm)		(Kg.cm²)	(mm)	(Kg)
MP086NA101	1.7	1.7	7.3	137.2	7.84	8	0.64	62	1.6
MP086NA102	1.25	4	16	137.2	7.84	8	0.64	62	1.6
MP086NA103	3.6	0.45	1.7	137.2	7.84	8	0.64	62	1.6
MP086NA104	4.5	0.31	1.1	137.2	7.84	8	0.64	62	1.6
MP086NA201	4	0.75	3.8	274.4	14.7	8	1.3	94	2.6
MP086NA202	4.6	0.55	2.5	274.4	14.7	8	1.3	94	2.6
MP086NA203	2.5	1.7	7.8	274.4	14.7	8	1.3	94	2.6
MP086NA204	2	2.7	11	274.4	14.7	8	1.3	94	2.6
MP086NA301	7	0.29	2	392	24.5	8	1.9	127	3.6
MP086NA302	4	1	5	392	24.5	8	1.9	127	3.6

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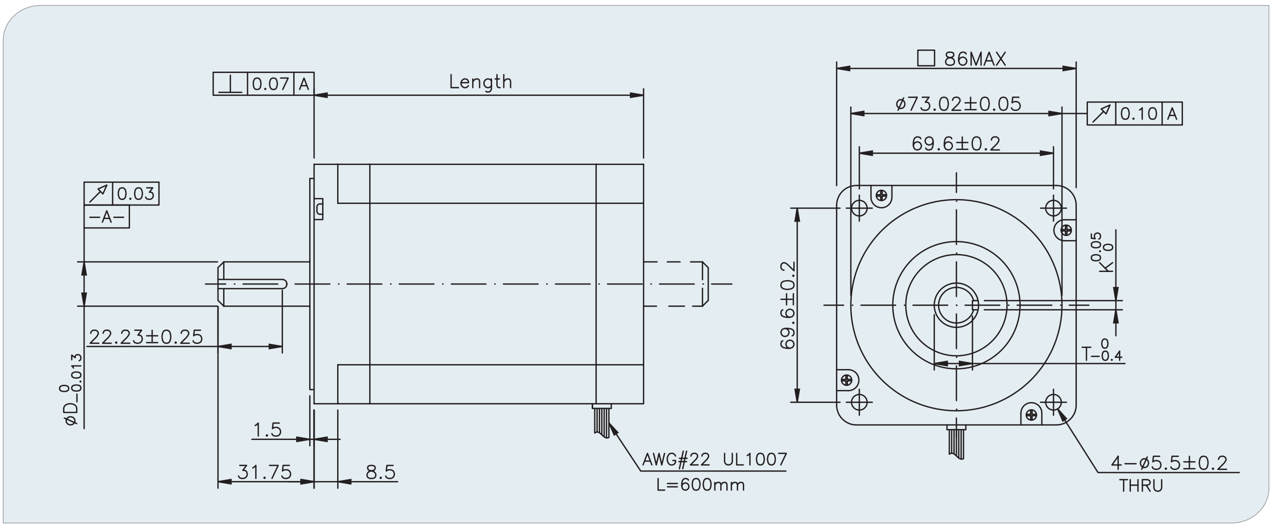
MP086YG

Stepper Motor

- 86mm, 1.8Degree Enhanced Hybrid Stepper Motor

Mechanical



Specifications

Part No.	Connection	Phase Current	Phase Resistance	Phase Inductance	Holding Torque	Detent Torque	No.of Wires	Rotor Inertia	Length	Weight	D	K	T
		(A)	(OHMS)	(mH)	(Nm)	(Nm)		(kg.cm²)	(mm)	(Kg)	(mm)	(mm)	(mm)
MP086YG100	Parallel	6.1	0.25	2.1	2.8	0.2	8	0.66	67	1.6	12.7	3.175	10.92
	Series	3	1	8.4	2.8	0.2	8	0.66	67	1.6	12.7	3.175	10.92
	Unipolar	4.3	0.5	2.1	2.3	0.2	8	0.66	67	1.6	12.7	3.175	10.92
MP086YG200	Parallel	8.6	0.18	1.4	4.59	0.23	8	1.4	79.5	2.27	12.7	3.175	10.92
	Series	4.3	0.72	5.8	4.59	0.23	8	1.4	79.5	2.27	12.7	3.175	10.92
	Unipolar	6.1	0.36	1.4	3.25	0.23	8	1.4	79.5	2.27	12.7	3.175	10.92
MP086YG300	Parallel	10	0.18	1.8	8.58	0.25	8	2.7	118	3.81	12.7	3.175	10.92
	Series	5	0.7	7	8.58	0.25	8	2.7	118	3.81	12.7	3.175	10.92
	Unipolar	7.1	0.35	1.8	6.07	0.25	8	2.7	118	3.81	12.7	3.175	10.92
MP086YG400	Parallel	9.9	0.22	2.3	12.1	0.38	8	4	156.5	5.39	15.87	4.763	13.13
	Series	5	0.87	9	12.1	0.38	8	4	156.5	5.39	15.87	4.763	13.13
	Unipolar	7	0.44	2.3	8.58	0.38	8	4	156.5	5.39	15.87	4.763	13.13

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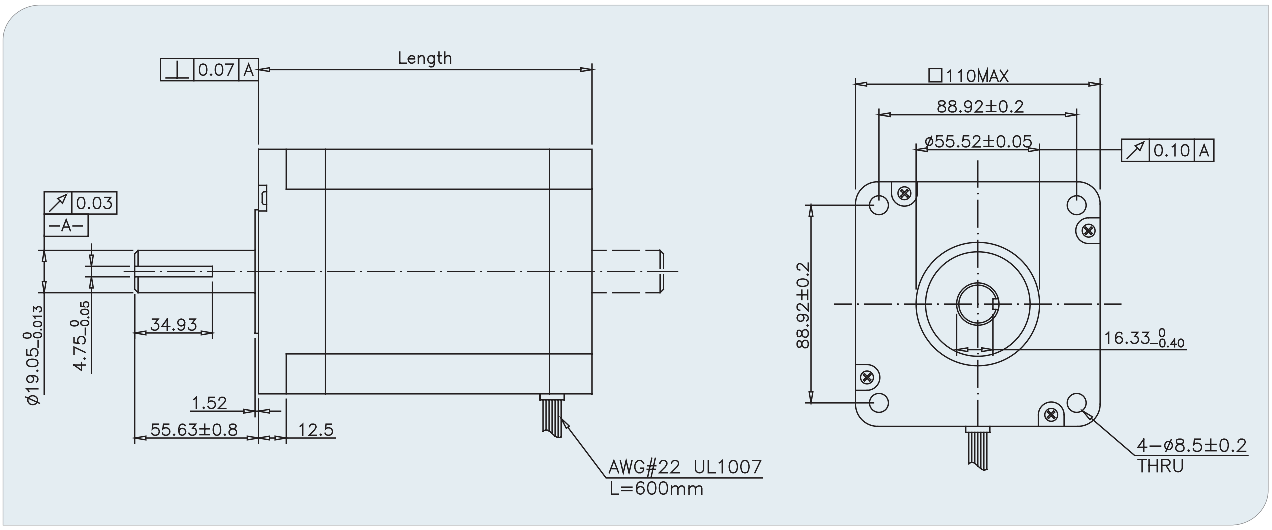
MP110YG
Stepper Motor

- 110mm, 1.8Degree Enhanced Hybrid Stepper Motor

Mechanical



Specifications

Part No.	Connection	Phase Current (A)	Phase Resistance (OHMS)	Phase Inductance (mH)	Holding Torque (Nm)	Detent Torque (Nm)	No.of Wires	Rotor Inertia (Kg.cm^2)	Length (mm)	Weight (Kg)
MP110YG100	Parallel	10.7	0.16	2.8	11.68	0.3	8	5.5	99	4.96
	Series	5.3	0.63	11.1	11.68	0.3	8	5.5	99	4.96
	Unipolar	7.5	0.31	2.8	8.26	0.3	8	5.5	99	4.96
MP110YG200	Parallel	15.8	0.1	2.1	22.09	0.59	8	10.9	150	8.34
	Series	7.9	0.41	8.4	22.09	0.59	8	10.9	150	8.34
	Unipolar	11.2	0.21	2.1	15.63	0.59	8	10.9	150	8.34
MP110YG300	Parallel	15.4	0.14	3.2	30.81	0.75	8	16.2	201	11.64
	Series	7.7	0.55	13	30.81	0.75	8	16.2	201	11.64
	Unipolar	10.9	0.28	3.2	21.81	0.75	8	16.2	201	11.64

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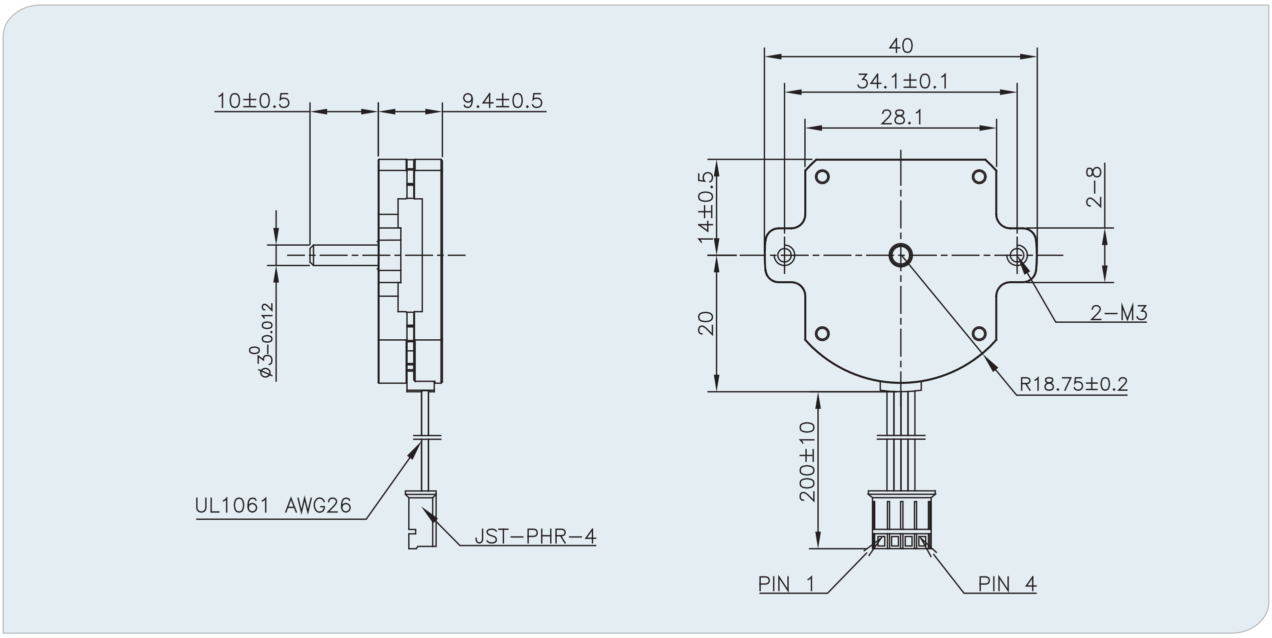
MPF028NB

Flat Hybrid Stepper

General information

- Step Angle: 1.8°±5%
- IP Code： IP 30
- Performance customization available

Dimension(Unit:mm)



Specifications

MODEL	UNIT	MPF028NB001
Nominal Voltage	Vdc	1.85
Current	A	0.5
Resistance	Ohms	3.7
Inductance	mH	0.88
Holding Torque	Nm	$9.8*10^{-3}$
Rotor Inertia	$Kg.m^2$	$1.7*10^{-7}$
Weight	Kg	0.028

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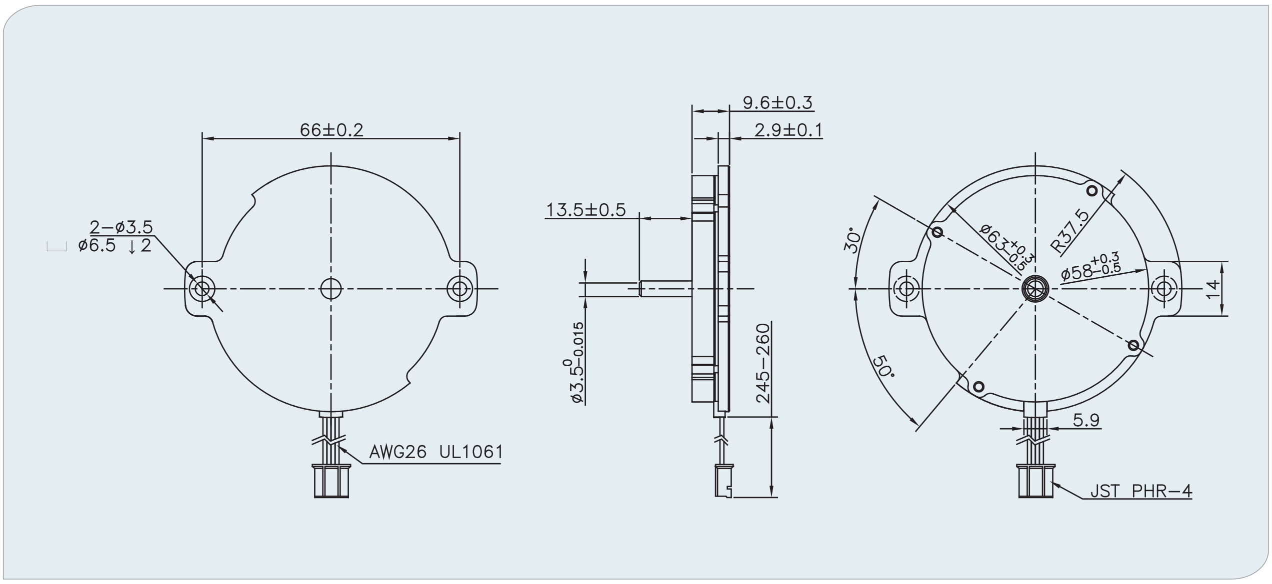
MPF068NB

Flat Hybrid Stepper

General information

- Step Angle: 1.8°±5%
- Ambient Temperature: -24°C - +60°C
- Performance customization available

Dimension(Unit:mm)



Specifications

MODEL	UNIT	MPF068NB001
Nominal Voltage	Vdc	3.8
Current	A	1
Resistance	Ohms	3.8
Inductance	mH	2
Holding Torque	Nm	0.064
Detent Torque	Nm	0.005
Rotor Inertia	$Kg.m^2$	$1.6*10^{-6}$
Weight	Kg	0.095

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